MIL-N-50071C
30 SEPTEMBER 1964
SUPERSEDING
MIL-N-50071B(CBRA)
13 SEPTEMBER 1962

MILITARY SPECIFICATION

NOSECUP, C10

This specification is mandatory for use by all Departments and Agencies of Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of protective mask nosecup.

1.2 Classification. Nosecups shall be of the following sizes, as specified (see 6.2):

Medium.
Large.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

MIL-STD-601 —Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 —Marking for Shipment and Storage.

DRAWINGS

EDGEWOOD ARSENAL

D5-2-970 —Nosecup, Medium.

D5-2-972 —Nosecup, Large.

PUBLICATIONS

EDGEWOOD ARSENAL DIRECTIVE

No. 618 – Ozone Resistance Test of Natural Rubber Material.

(Copies of specifications, standards, drawings, and publications required by suppliers, in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 4240

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Rubber compound.* The rubber compound shall be nonblooming, except for slight wax-type bloom, and shall contain not less than 65 percent, by volume, of new natural rubber. The compound shall be free from lead or mineral sulfates.

3.1.2 *Antioxidants and accelerators.* The rubber compound shall contain only approved antioxidants and accelerators (see 3.1.6).

3.1.3 *Antiozonide.* The rubber compound shall contain an antiozonide.

3.1.4 *Copper and manganese inhibitor.* The rubber compound shall contain a copper and a manganese inhibitor.

3.1.5 *Coloring materials.* The rubber compound shall be colored by carbon black as required.

3.1.6 *Toxicity.* The nosecup shall not cause undue primary skin irritation or sensitization of the skin when assembled and worn as a component of protective masks. Objective evidence shall be submitted in accordance with 4.2.2.2. One hundred and twenty-five masks and thirty test slabs, having the same formulation and equivalent cure (see 6.3) as will be used during regular production, shall be submitted to the Government for testing as specified in 4.7.3, except when the formulation has been approved prior to preproduction. Thirty additional test slabs shall be submitted for approval under the conditions cited above, whenever a major change is made (see 6.4). The supplier shall not make a major change in rubber formulation or the curing process after receiving approval of a formulation from the Government.

3.2 Manufacture. The nosecup shall be manufactured, in one piece, by the conventional process of molding under heat and pressure. Each size specified in the contract or order shall be manufactured in accordance with the applicable drawing listed in 2.1 in molds approved by the Government.

3.2.1 *Repair of flaws.* Nosecups with small flaws may be repaired and accepted provided there is no change in the formulation or surface continuity.

3.2.2 *Test slabs.* Test slabs, 6 by 6 by 0.075 ± 0.01 inches, shall be made from stock representative of the formulation used for the preproduction sample and from each sublot during regular production of nosecups. The slabs shall have a cure equivalent to that of the regular production nosecups (see 6.3).

3.3 Physical properties. Physical properties shall conform to the requirements shown in table I when tested as specified therein.

3.4 Preproduction sample. Prior to the start of regular production, fifteen (10 trimmed and 5 untrimmed) sample nosecups from each mold cavity and 39 test slabs shall be produced using the same methods, materials, equipment, and rubber formulation as will be used during regular production. Any change in formulation, operating conditions, or molds shall require a new preproduction sample.

3.5 Critical areas. The peripheral sealing area of the valve and the flange surfaces shall be considered as critical areas of the nosecup.

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3.6 **Workmanship.** The nosecup inner and outer surfaces shall be smooth and free from imperfections, such as pits, bumps, porous areas, cracks, checks, flash, abrasions, blisters, or foreign matter. The rubber shall be compact and free from holes and bubbles. Ridges of mold parting lines on outer surfaces shall not be considered as flash when closely trimmed. Rips, tears, or punctures shall be considered as damage. Workmanship defects of a magnitude greater than the referee standards established by the Government for major A, major, and minor defects (see 6.8) shall be categorized in the appropriate defect class.

TABLE I. *Physical Properties*

Property	Unaged	After heat aging	Test paragraph
Average tensile strength, psi, min	3000	2550	4.7.2.1
Average ultimate elongation, %, min	600		4.7.2.1
Percent increase, max		20	
Percent decrease, max		25	
Average tensile stress at 200% elongation, psi	250 to 450	250 to 450	4.7.2.1
Individuals, psi, min	240	240	
Average tear resistance, lbs/in, min			4.7.2.1
Die A or B	300	210	
Die C	225	165	
Average tension set at 75% ultimate elongation, % max	20		4.7.2.1
Average hardness (Shore "A" or "A2")	47 ± 3		4.7.2.1
Ozone resistance, min	80		4.7.2.2
Flexibility:			
Units at 75° ± 5°F., min	7.5		
Units retained at — 25° ± 5°F., min	4.8		
Puncture resistance, psi, min	9.5		4.7.2.4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 *Supplier's responsibility.* Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to specified requirements.

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4.1.2 *Government's responsibility.* The Government will be responsible for the performance of the inspection set forth in 4.7.3. Samples shall be forwarded to the laboratory designated by the contracting officer.

4.2 Special provisions.

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.*

4.2.2.1 *Materials.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.2.2 *Toxicity.* The supplier shall submit to the contracting officer within 30 days after date of award of a contract, at no additional cost to the Government, a complete list of the ingredients to be used in the rubber formulation, the percentage of each, the cure time and temperature to be used, and the tolerances of the above that will be permitted as well as the mold lubricants, dusting powders, or wet sprays to be used. The supplier shall also submit with the above list their physical test results and complete evidence substantiating his claim, to the satisfaction of the Government, that the listed components will be nonirritating or nonsensitizing to the wearer of the end item involved. This evidence may be in the form of a complete description of the programs that the supplier has carried out, such as patch testing or use testing. He may submit evidence that the proposed formulation is essentially the same as that used in his commercial products, and include information relative to the extent and length of time of his commercial experience with the formulation. Other evidence may include reference to technical or other scientific publications, laboratory reports, opinions or statements of recognized authorities or agencies in the field of dermatology, or other pertinent data. The Surgeon General will act as advisor in evaluating this evidence for determining if the components are acceptable (see 4.7.3).

4.2.3 *Inspection substation.* If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL-STD-105 are applicable to each substation and to the end item.

4.3 Preproduction sample inspection.

4.3.1 *Examination.* The sample nosecups shall be inspected for all requirements of the drawings and this specification.

4.3.2 *Tests.* Test slabs produced in accordance with 3.4 shall be tested in accordance with 4.7.2. Nosecup material produced in accordance with 3.1.6 shall be tested in accordance with 4.7.3, except where the toxicity formulation has been approved prior to preproduction.

4.3.3 *Acceptance/rejection criteria.* The preproduction sample shall meet the examination and tests specified in 4.3.1 and 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

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4.4 **Lotting.** A regular production lot shall consist of the nosecups of one size produced in sequence during one month as indicated by the dots in the medallion of the nosecup. A sublot shall consist of the nosecups produced from one batch of material (see 6.5). A new lot shall be established following the occurrence of any event affecting the homogeneity of the lot or use of an alternate approved formulation. Acceptably repaired nosecups from any accepted parent lot may be incorporated into and assume the lot number of current production.

4.5 **Sampling.**

4.5.1 *For examination.* Sampling shall be conducted in accordance with MIL-STD-105.

4.5.2 *For destructive tests.* Thirty-nine test slabs shall be submitted with each sublot for testing.

4.6 **Inspection provisions.**

4.6.1 *For examination.* The sample nosecups from the regular production lot shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.6.2 *For destructive tests.* Sample test slabs shall be tested in accordance with 4.7.2. The acceptance number is 0.

4.6.3 *Classification of defects.*

4.6.3.1 *Nosecups.* (See applicable dwg for size.)

Categories	*Defects*
Critical:	None defined.
Major:	AQL 1.0 percent defective.
101	Thickness.
102	Distortion of deformation.
103	Workmanship (other than Major A 150, see 3.6).
104	Marking incorrect.
105	Button holes incorrect or incorrectly located.
Minor:	AQL 2.5 percent defective.
201	Marking missing or illegible.
202	Excessive flash or improper trim.
203	Size incorrect.
Major A:	AQL 0.065 percent defective.
150	Imperfections or damage in critical areas.

4.7 **Tests.**

4.7.1 *Test specimens.* Specimens shall not be buffed or stretched before testing. Twenty specimens, two from each of 10 test slabs, shall be taken for tensile strength, ultimate elongation, and tensile stress testing. Ten specimens shall be tested before aging and 10 tested after aging. Six specimens, two from each of three test slabs, shall be taken, without aging, for hardness and tension set testing. Twenty specimens, two from each of 10 test slabs, shall be taken for tear resistance testing. Ten specimens shall be tested before aging and 10 tested after aging. Twelve specimens, two from each of six test slabs, shall be taken for ozone resistance testing. Eight specimens, two from each of four test slabs, shall be taken for flexibility testing. Six specimens, one from each of six test slabs, shall be taken for puncture resistance testing. The test specimens selected for heat aging shall be aged in an air oven at a temperature of 158° ± 1.8°F. for a continuous period of 168 ± 1 hours or at a temperature of 212° ± 1.8°F. for 24 ± 1 continuous hours, as specified in FED-STD-601 (see 6.6).

4.7.2 *Physical properties test.*

4.7.2.1 *Tensile strength, ultimate elongation, tensile stress, tear resistance, tension set, and hardness.* Tensile strength, ultimate elongation, tensile stress, tear resistance, tension set, and hardness shall be determined

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as specified in FED-STD-601. The angle-type or crescent-type die shall be used for tear resistance testing.

4.7.2.2 *Ozone resistance.* Ozone resistance shall be determined in accordance with Directive No. 618.

4.7.2.3 *Flexibility.* Flexibility shall be determined using Edgewood Arsenal Q37 Natural Rubber Flexibility Tester. Flexibility shall be read on the grid of the apparatus. Four specimens shall be tested at 75° ± 5°F. and four specimens shall be tested at −25° ± 5°F. after being conditioned for 72 hours at that temperature.

47.2.4 *Puncture resistance.* Resistance to hydrostatic pressure shall be determined using Edgewood Arsenal Q38 Natural Rubber Puncture Tester.

4.7.3 *Toxicity.* All masks and test slabs submitted to the Government for testing are subject to evaluation from the toxicity stand point by the Office of the Surgeon General. Such tests as are deemed necessary by the Office of the Surgeon General for properly evaluating the toxicity of any protective mask or test slab will be carried out in accordance with direction of the Surgeon General.

5. PREPARATION FOR DELIVERY (For repair parts packaging, see 6.7).

5.1 Packaging, level C. Nosecups of one size shall be packaged to afford adequate protection and to prevent any deformation or damage during shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Nosecups of one size, packaged as specified in 5.1, shall be packed in such manner as to insure carrier acceptance and safe delivery at first destination for immediate use. Containers shall comply with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129. The following additional marking shall appear on all unit packages and shipping containers:

(a) Nosecup size.
(b) Composition of nosecups and formulation code.
(c) Nomenclature of mask to which nosecup pertains.
(d) Date of cure.

6. NOTES

6.1 Intended use. The nosecup covered by this specification is intended for use in the manufacture of ABC-M17 Field Protective Masks.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Size of nosecup.
(c) Applicable drawings.

6.3 Equivalent cure. Slabs and nosecups of the same stock and gage are considered to have equivalent cures or the same state of vulcanization when:

(a) Their free-sulfur contents do not vary more than 10 percent.
(b) Their average tensile-stress values at any given elongation (Other than ultimate elongation) do not vary more than ± 2.5 percent.
(c) Their curing times can be equated in the following equation:

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$$t_2 = t_1 \times 1.6^{\frac{T_1 - T_2}{10}}$$

where: t_1 = minutes of cure at temperature T_1,

t_2 = minutes of cure at temperature T_2,

T_1 = temperature (°F.) of cure for time t_1, and

T_2 = temperature (°F.) of cure for time t_2.

For convenience, the nomograph in figure 1 may be used instead of the mathematical computation time-temperature relationships in determining equivalent cure.

6.4 Major change. The term applies to any change in formulae, processing, or curing of currently approved rubber stocks which may induce toxic characteristics and cause casualties. The major change should be considered effected when one or more of the following conditions occur:

(a) Components of formulae are varied in excess of the limitations stated in the manufacturer's tolerance (see 4.2.2.2).

(b) The addition or deletion of one or more components of formulae.

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EQUIVALENT CURE TIME

FOR

NATURAL RUBBER

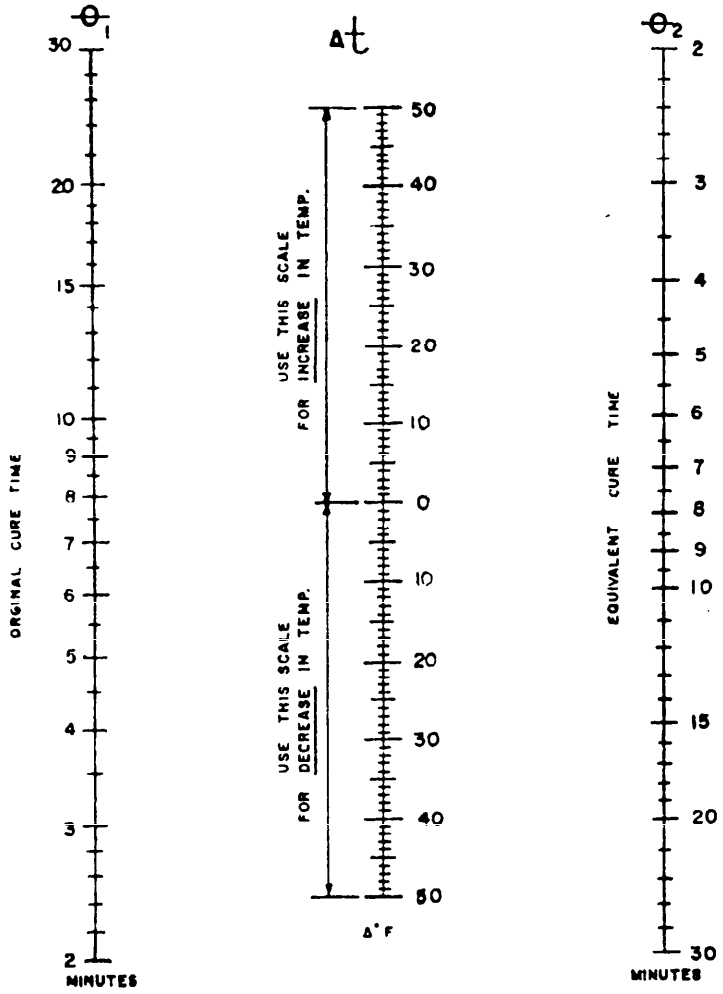


Figure 1.

USING A STRAIGHT EDGE, CONNECT THE ORIGINAL CURE TIME ON θ_1 SCALE WITH THE CHANGE OF CURE TEMPERATURE ON Δt SCALE. EXTEND THE LINE UNTIL IT INTERSECTS θ_2 SCALE AT THE EQUIVALENT CURE TIME.

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(c) Substitutes for one or more components of formulae without advance approval of the Government.

(d) Deviations from the submitted and approved factory mold cure exceeding + 5°F. or −10°F., even though equivalent cure is attained.

(e) Use of mold lubricants, dusting powders, or wet sprays not used in preparation or cure of the original samples submitted for toxicity approval.

6.5 **Batch.** A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.6 **Aging.** Aging at 212° ± 1.8°F. for a continuous period of 24 hours is slightly more severe on rubber than aging at 158° ± 1.8°F. for 168 consecutive hours; therefore, specimens that fail to meet requirements when aged at the higher temperature may be discarded and replaced by new specimens aged at 158° ± 1.8°F. for 168 consecutive hours.

6.7 **Repair parts packaging.** When repair parts packaging and packing is required for items authorized for stocking as repair parts, the repair parts packaging data sheet covering the C10 Nosecup should be used.

(Copies of repair parts packaging data sheets may be obtained from the Director for Engineering and Industrial Services, ATTN: Chief, Design Branch, US Army Edgewood Arsenal, Md. 21010.)

6.8 **Referee standards.** During preproduction and subject to later revision during production, the Government will establish referee standards to define the type and category (Major, Minor, and Major A) of each defect. This classification will be based upon the defect magnitude and potential deliterious effect on intended use.

6.9 **International interest.** Certain provisions of this specification are the subject of ABC international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:

Army—MU

Navy—SH

Air Force—84

Preparing activity:

Army—MU(EA)

Project No. 4240-0069

International interest: (see 6.9)

Review activities:

Army—MD

Navy—SH

User activities:

Navy—MC

Air Force—84

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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PLATE NO. 15419 (BA

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The 125 sample masks of each size